

# Calibrated Uncertainty for Active Room-Layout Estimation

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**Abstract**—A robot that estimates the layout of a room usually reports geometry and little else. Downstream consumers—planners, scene graphs, other agents—must then decide how far to trust it, with nothing to go on. We present an online room-layout estimator whose reported uncertainty is calibrated against ground truth, and which uses that uncertainty as a decision variable: it gates which corners are published, and it is what the robot acts to reduce. Parameters and structure (how many walls there are) are inferred by minimising a variational free energy; viewpoint selection minimises the corresponding expected free energy, whose epistemic term reduces to expected information gain under the Gaussian model. The shared currency is what lets the data fit, the model complexity and the value of the next observation be compared at all, and the calibrated uncertainty is what the robot acts on. On 594 annotated floor plans the estimator reaches 0.983 mean IoU, and—our central result—a corner it declares publishable has a median true error of 2.0 cm and lies within its declared 6 cm in 86 % of cases, against 3.6 cm and 64 % for corners it withholds. Holding the system to that standard proved diagnostic: it exposed three distinct precision defects in three different modules, all instances of uncertainty that averages away past an error that does not, one of them in a module the instrument was not measuring. A single global precision is insufficient for this problem: the part of the error that does not average is several times larger than the part that does, it depends on the viewpoint rather than on the map, and the precision declared for it is smaller than it—so a constant calibrated under one condition need not remain calibrated under another, and precision should be inferred rather than tuned.

## I. INTRODUCTION

Room-layout estimation is mature enough that accuracy alone no longer separates methods. What separates them for a robot is whether the estimate can be *acted on*: whether a planner can tell a wall that is known to a centimetre from one that is a guess, whether a scene graph should commit a corner to shared memory, whether the robot should look again before deciding. All of these are questions about uncertainty, and almost no layout system answers them with a number that has been checked.

This is not an oversight of presentation. An uncertainty that is never validated tends to be wrong, and wrong in a way that is invisible: an estimator that overstates its error is merely timid, while one that understates it is confidently incorrect, and neither shows up in a mean IoU. We found both failure modes in our own system, and we found them only because something downstream was made to depend on the number being right.

Our estimator maintains a room as a cyclic polygon of wall landmarks in Hesse normal form, inferred jointly with the robot pose over a sliding window. Three decisions that are

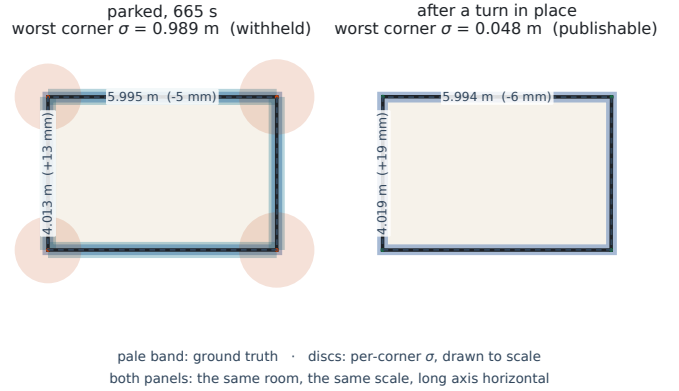


Fig. 1. **The estimate is right before it is confident, and says so.** One closed-loop run; the same room in both panels, same scale, long axis horizontal. Pale band: ground truth, each dimension carrying its measured value and signed error. Discs: per-corner  $\sigma$ , to scale. *Left*, parked for 665 s: the geometry is already within 13 mm of truth, yet every corner declares almost a metre of uncertainty and the layout is withheld. *Right*, the first publishable frame, after a turn in place: the geometry moves by at most 6 mm, and stays within 19 mm of truth, while corner  $\sigma$  falls by a factor of 21. The estimator does not earn precision by waiting, and does not claim any.

usually taken by three separate mechanisms are here taken by one. *What the walls are* is variational inference over their parameters. *How many walls there are* is Bayesian model comparison: a structural change is adopted when it pays for its own description length, priced in nats on the same scale as the data fit, rather than by a merge threshold. *Where to look next* is the epistemic term of the expected free energy, so the robot moves to reduce the uncertainty that is currently blocking it from committing to a layout. Because all three are denominated in the same currency, every precision in the model carries weight: get one wrong and the comparisons it feeds are priced in a corrupt unit. That is the property we exploit.

### Contributions.

- 1) An online layout estimator whose per-corner and per-edge uncertainty is *validated against ground truth* on 594 annotated floor plans, and is *actionable*: it gates publication and is the quantity the explorer acts to reduce (§V). We also report where that buys nothing: against a heuristic explorer on the same corpus the difference is null, and we say why.
- 2) A propagation from returns to corner covariance that

separates the error that averages down with more observation from the error that does not, and a measurement of both terms on real sensor data (§V-C).

- 3) Evidence that a single global precision is insufficient for this problem: the non-averaging part of the error is several times larger than the averaging part, it is viewpoint-dependent rather than a property of the map, and the declared pose precision does not cover it—measured in simulation, with the size of that bias reproduced on a physical robot (§V-E).
- 4) Three precision defects found by calibration, including one localised in a module the statistic was not measuring—offered as evidence that a free-energy formulation is useful diagnostically, not only descriptively.

## II. RELATED WORK

*Floor-plan reconstruction from complete scans.* A large family of methods recovers a floor plan from an accumulated point cloud or density image, from the early unified formulation [1] and its shortest-path successor [2] through attention and query-based architectures [3], [4], room-wise implicit and slicing representations [5], [6], room-aware transformers [7], continuity-aware edges [8], guided set diffusion [9], autoregressive polygon sequences [10], and segmentation-guided fusion [11]. They are evaluated by corner and room F1 on a finished scan. Two properties distinguish them from our setting: they are offline, so no decision is taken while the estimate is still uncertain; and the number of corners is chosen by a search with hand-set costs rather than by a likelihood, so there is no quantity in which structure and data fit are commensurate.

*Panoramic room layout.* Single- and multi-view layout estimation recovers room shape from 360° imagery [12], [13], [14], with self-training and multi-view consistency removing the need for per-scene labels [15], [16], and related work registering panoramas or verifying their alignment [17], [18]. These produce a layout per view rather than a belief a robot accumulates, and report no per-corner uncertainty a consumer could gate on. Unbounded and general (non-Manhattan) variants relax the shape prior [19], [20], [21]. *Sequential* floor-plan estimation from a moving camera [22] is the closest in spirit to ours: it, too, must decide room membership and shape while the scene is still incomplete. Autoregressive scene description [23] likewise produces structure from a walkthrough, as language rather than as a belief with a covariance.

*Exploration that predicts structure.* A complementary line predicts unobserved layout to drive exploration—room shape from partial observation [24], walls beyond the frontier learned from floor plans [25], information gain from global map predictions [26], and generative layout prediction for planning [27]—together with the question of when a map is finished [28]. These share our premise that a layout belief should inform where to go next; the predicted structure is used to *score* viewpoints rather than to be published with an uncertainty a downstream consumer must trust.

*Active SLAM.* Surveys of the field [29], [30] formalise viewpoint selection by information-theoretic utility, and the question of whether the usual uncertainty criteria are the right ones has itself been raised [31]. Our selection rule is the epistemic term of an expected free energy, which coincides with information gain under a Gaussian model; the contribution here is not the rule but that the precision it consumes has been verified, so the gain it predicts is a gain that exists.

*Uncertainty and calibration.* Uncertainty estimation is well studied in dense depth prediction, where predicted confidence can be compared against realised error [32], and in photogrammetric modelling [33], [34]. To our knowledge no online room-layout estimator reports a per-corner covariance that has been validated against layout ground truth, which is the gap this paper addresses. Our position is deliberately parametric: we keep a covariance rather than a coverage set, because a joint estimator and a downstream consumer both need one, and we check it.

*What the two literatures leave open.* Layout methods are evaluated on geometric accuracy; none of them calibrates a *per-corner* uncertainty against the error it actually makes, so a consumer is given a shape and no basis for trusting any part of it more than another. Active SLAM and next-best-view methods do consume a covariance, but they inherit whichever one the estimator produces and rarely ask whether that covariance predicts the realised error — an acquisition policy driven by an over-confident covariance will stop moving precisely where it should not. The two gaps are the same gap seen from opposite sides. Our contribution joins them: we calibrate the layout’s uncertainty against ground truth, and then use the calibrated quantity for both jobs it is fit for — as the criterion that decides which geometry may be published, and as the objective the robot moves to reduce.

## III. GENERATIVE MODEL

A room is a cyclic sequence of wall landmarks. Wall  $w$  is a line in Hesse normal form,  $\mathbf{n}(\phi_w) \cdot \mathbf{p} = d_w$ , with belief  $\mathcal{N}([\phi_w, d_w]^\top, \Lambda_w^{-1})$ . The robot pose over a sliding window of  $K$  frames is  $x_{1:K}$ . Given returns  $z$ , perception minimises the variational free energy

$$F = \underbrace{-\log p(z \mid x, \Phi)}_{\text{accuracy}} + \underbrace{D_{\text{KL}}(q(x, \Phi) \parallel p(x, \Phi))}_{\text{complexity}}, \quad (1)$$

which under a Gaussian  $q$  reduces to the sparse nonlinear least-squares problem we solve by Gauss–Newton over the window.

*Structure as model comparison.* The number of walls is not fixed. A candidate structural change—birth, death, or the splice of a new edge into the cycle—is a different model  $m'$ , and is adopted when  $F(m') < F(m)$ . The complexity term charges each additional edge a fixed description length: with a room of radius  $R$  discretised at resolution  $c$ , an edge costs  $\ln 4 + \ln(2R/c)$  nats to name. There is no merge threshold and no minimum wall length; a thin partition survives exactly when the returns that support it pay for the nats it costs.

*Action.* A viewpoint  $v$  is scored by the expected information gain  $\mathbb{E}_o[H(x, \Phi) - H(x, \Phi \mid o)]$ , predicted through the

model’s own forward sensor: beams are cast against the current belief, and each wall’s expected Fisher contribution  $J_w(v)$  gives a Gaussian gain  $\frac{1}{2} \ln(\det(\Lambda_w + J_w)/\det \Lambda_w)$ . Because publication is gated on corner  $\sigma$  (§IV), the robot acts, in effect, to make its own map publishable.

#### IV. WHAT A LAYOUT ESTIMATE IS WORTH

A corner is the intersection of two walls, so its covariance follows from theirs. With  $M = [\mathbf{n}_a^\top; \mathbf{n}_b^\top]$  and the intersection  $\mathbf{p}$ ,

$$\Sigma_{\mathbf{p}} = J_a \Lambda_a^{-1} J_a^\top + J_b \Lambda_b^{-1} J_b^\top + \sigma_m^2 I, \quad (2)$$

where  $J$ ’s  $\phi$  column carries the lever arm  $\mathbf{t} \cdot \mathbf{p}$  and  $M^{-1}$  contributes  $1/\sin \theta$  between the walls, so a shallow corner is correctly reported as uncertain along its bisector without any angle test.

The term that matters for this paper is the last one. The first two shrink as  $1/N$  with the returns behind each wall: observe longer and they go to zero. *The true error does not follow them down.* What remains when the statistics are exhausted is systematic—the offset between the mapped wall and the real surface, identical on every look from a given viewpoint—and it is what  $\sigma_m$  represents. Omitting it produces an estimator that grows arbitrarily confident about a corner whose error is bounded below; double-counting it produces one that is timid. We measure it rather than tune it, and §V-E shows why a constant does not suffice.

#### V. EXPERIMENTS

We evaluate at three levels, and they establish different things. §V-A tests geometric inference and uncertainty calibration on 594 real room shapes, with observations simulated from the annotations. §V-C validates the noise decomposition itself on real Matterport depth, which is what licenses the simulated sensor used in the first level. §V-D runs the estimator online in simulation with a LiDAR sensor model, where the uncertainty gates publication as it is produced, and §V-E closes with the same measurement repeated on a physical robot. No *closed-loop* experiment runs on hardware—the layout accuracy, the calibration and the closed loop are all simulated—and we are explicit throughout about which claim rests on which level.

##### A. Geometric generality: 594 annotated floor plans

We evaluate on 594 room annotations from MatterportLayout [35], the Manhattan room-layout annotations on Matterport3D [36] panoramas, spanning 6 to 18 ground-truth corners.

*What this measures.* The observations are *simulated from the annotated polygon*: a range sensor is driven through the annotated room and the estimator never sees a photograph. Because the returns are synthesised from the annotated polygons, this experiment evaluates geometric inference and uncertainty calibration rather than appearance-based layout recognition, and Table I is not comparable with single-image methods evaluated on the same corpus. What the 594 rooms do provide is the *shape distribution of real buildings*: real aspect ratios, real sizes, and non-convex plans with up to 18 corners, in place

TABLE I  
LAYOUT ACCURACY, 594 ANNOTATED FLOOR PLANS, ONE RUN.

|                              |         |
|------------------------------|---------|
| IoU, mean                    | 0.9827  |
| IoU, median                  | 0.9870  |
| IoU, 10th percentile         | 0.9690  |
| IoU $\geq 0.95$              | 95.2 %  |
| IoU $\geq 0.90$              | 98.8 %  |
| Failures (no closed polygon) | 8 / 594 |

of a generator’s idea of a room. The quantity under test here is the estimator’s geometry and, in §V-B, its uncertainty—for which the annotated corners are ground truth regardless of how the returns were produced. §V-C then validates the sensor model itself on real depth.

*Does the epistemic term earn its place?* The explorer scores viewpoints by expected information gain. To test whether that objective is what produces the accuracy above, we re-ran all 594 rooms with the epistemic term replaced by the frontier-and-coverage heuristic it superseded, holding the estimator, the sensor model, the per-room seeds and the 900-frame budget fixed. *We find no statistically detectable difference under this evaluation.* Paired over the 583 rooms both arms complete, the mean IoU difference is  $-0.0007$  (95% CI  $[-0.0019, +0.0004]$ , bootstrap over rooms), the heuristic winning 263 rooms to 251; hard failures are 8 against 6; and the median frame at which the map first becomes publishable differs by 6 frames (95% CI  $[-14, +28]$ ). Splitting by ground-truth corner count recovers no difference in any band.

We report this because it bounds our own claim, and because the reason for the null is instructive. Every room in this corpus ran to the frame cap in both arms: the room sizes are such that 900 frames of any reasonable policy eventually sees every wall, so the *final* map cannot separate the policies, and an objective cannot be graded on an endpoint its own budget has saturated. What the experiment supports is that the uncertainty is consumed—it gates publication, and the explorer reads it. What it rules out is a claim we might otherwise have made: on rooms of this size, the epistemic objective buys no measurable accuracy or speed over a simple heuristic. Whether it pays under a tighter budget, or in the larger multi-room environments where a policy has room to waste time, is not tested here.

##### B. Is the reported uncertainty calibrated?

For every room we take the final published polygon and match its corners one-to-one to the truth corners by Hungarian assignment. Surplus corners on either side remain unmatched and are counted as *structure* errors—never as position error. This matters: nearest-vertex matching instead yields a mean error of 0.43 m against a median of 0.023 m, because a spurious corner has no truth counterpart and its “error” is meaningless.

Table II shows the effect of the systematic term of (2). Without it, the true error is flat at about 2 cm across every band of declared  $\sigma$ : a corner claiming 1 cm is no more accurate than one claiming 5 cm, and the declared uncertainty ranks corners

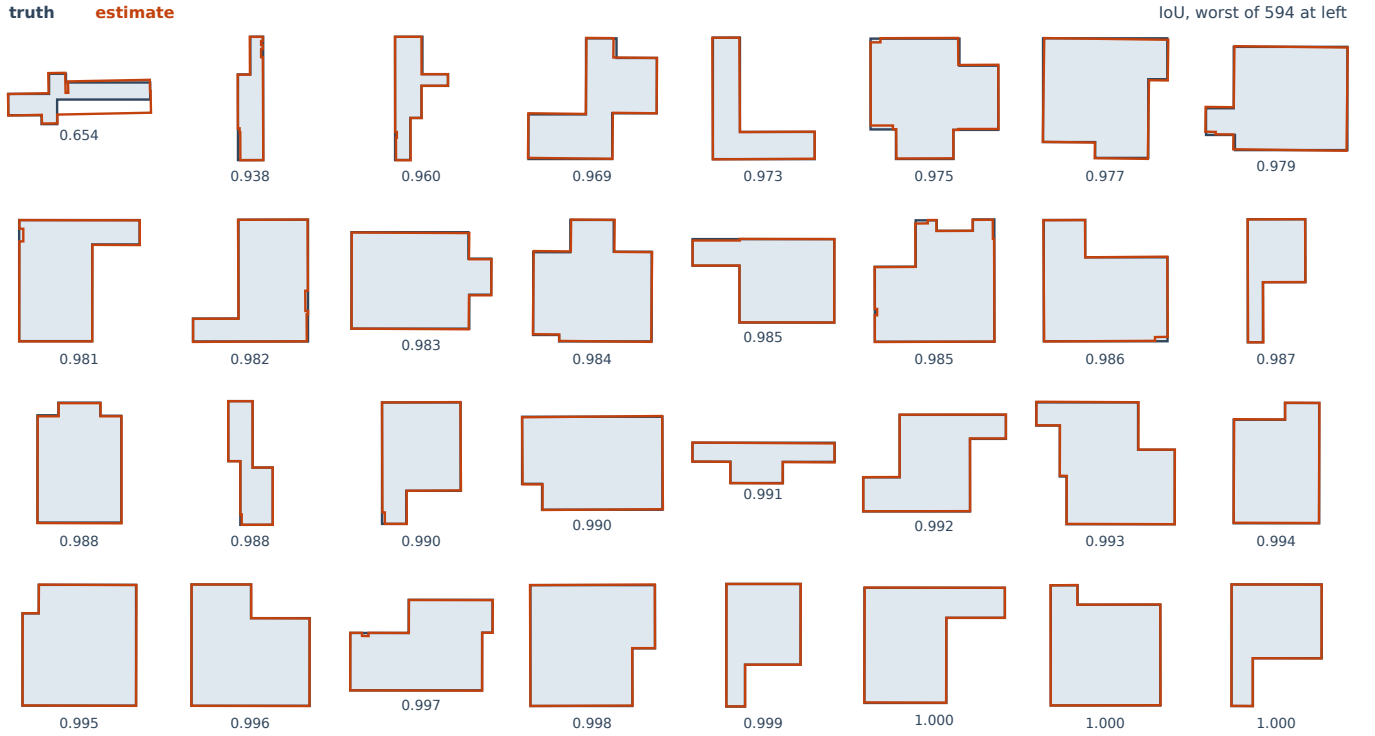


Fig. 2. **Thirty-two of the 594 annotated floor plans**, truth filled, estimate outlined, with the achieved IoU beneath each. The panels are ordered by IoU and sampled evenly across *every* room that produced a closed polygon, so the population’s worst room—IoU 0.654—is the first cell and the sample is not a selection of successes. The corpus’s own shapes run from 6 to 18 ground-truth corners, with the alcoves, wall columns and re-entrant corners that no rectangle prior would admit.

TABLE II  
CALIBRATION OF THE PUBLISHED CORNER  $\sigma$

|                            | $\leq 1\sigma$ | $\leq 2\sigma$ | $\leq 3\sigma$ | publish      | holds        |
|----------------------------|----------------|----------------|----------------|--------------|--------------|
| no $\sigma_m$              | 46.7%          | 70.5%          | 81.2%          | 89.0%        | 85.8%        |
| $\sigma_m = 2$ cm          | <b>65.3%</b>   | <b>85.0%</b>   | <b>91.2%</b>   | 89.0%        | 86.1%        |
| 95% CI, by room            | [62.7, 67.9]   | [83.2, 86.8]   | [89.9, 92.4]   | [87.8, 90.4] | [84.2, 87.8] |
| global $\sigma_c$ (fitted) | 65.3%          | —              | —              | 100%         | —            |

4344 matched corners, from the 586 of 594 rooms that closed a polygon — the other 8 have no corners to match. *publish* is the fraction passing the 6 cm gate, *holds* the fraction whose claim is borne out. The first three columns are empirical coverage,  $P(\|e\|_2 \leq k\sigma)$  — *not* a bound, since a standard deviation makes no containment guarantee. Intervals are 2000-sample bootstraps resampling *rooms*, not corners, which share a room’s walls and trajectory. The last row is a deliberately favourable baseline: one global  $\sigma_c$  fitted *post hoc* on this same corpus so its  $1\sigma$  coverage matches ours exactly.

without predicting their error. The measured floor is 0.019 m, against a simulated sensor of 0.020 m per return. Adding it as a variance term—not a clamp—brings the declared  $\sigma$  into agreement with the error that occurs, at almost no cost in yield.

*One global precision cannot do this job.* The last row of Table II is the obvious alternative: a single  $\sigma_c$  fitted to this very corpus, chosen so that it reproduces our  $1\sigma$  coverage exactly ( $\sigma_c = 0.0317$  m). It is therefore calibrated as favourably as the data allow — and it is still insufficient for the task. Being constant, it provides no ranking of corners by realised error at all: a rank correlation is not merely zero but undefined,

since the predictor has no rank variation. Under the 6 cm gate it publishes 100% of corners, because every corner carries the same number and the gate has nothing to separate. The post-hoc fit is not what defeats it: refitting  $\sigma_c$  by 5-fold cross-validation over *rooms* gives  $0.0318 \text{ m} \pm 0.0002 \text{ m}$  and a held-out  $1\sigma$  coverage of 65.4%, indistinguishable from the 65.3% it was fitted to reach. The scalar is perfectly well calibrated *on average* and generalises as such; what it cannot do is vary. A scalar can be calibrated *on average* and still be unable to tell one corner from another, which is the distinction this paper is about.

*What  $\sigma$  is, and is not, good for.* Between the two gate classes the separation is clean and survives the room-level interval: published corners have a median error of 0.0202 m (CI 0.0185 m to 0.0220 m) against 0.0364 m (CI 0.0312 m to 0.0446 m) for withheld ones, and the intervals do not overlap. Ranking *within* a class is weaker: the Spearman correlation between declared  $\sigma$  and realised error is 0.125 (CI 0.088 to 0.161)—significantly positive, but small. We report it because it bounds the claim honestly: this  $\sigma$  is a trustworthy *decision variable* for the publish/withhold question it is used for, not a fine-grained ordering of corners by accuracy.

*The gate is a tolerance, not a tuned parameter.* The 6 cm gate is the downstream consumer’s requirement—the clearance a planner in this stack needs a corner to be good to—and was fixed before this evaluation rather than chosen on it. Nothing

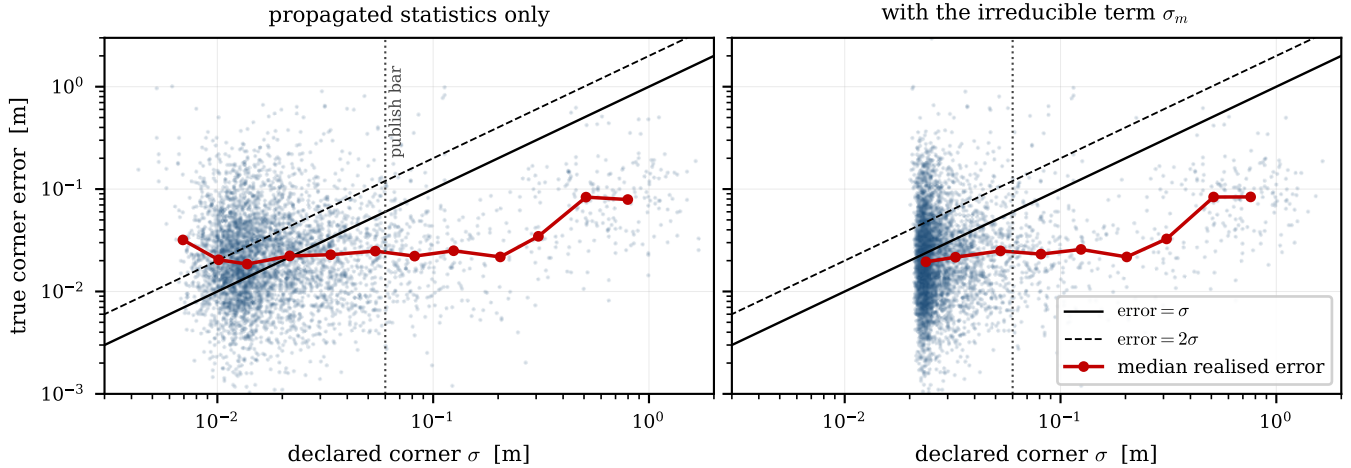


Fig. 3. **Does the declared uncertainty track the realised error?** All 4344 matched corners from the 586 annotated floor plans that produced a closed polygon: declared  $\sigma$  against the error it actually made. *Left*, propagating only the walls’ statistical uncertainty: the red median is flat, so a corner claiming 1 cm is no more accurate than one claiming 5 cm, and the cloud sits far above error =  $\sigma$  at the tight end. *Right*, with the irreducible term of (2): the median tracks the diagonal, so the declared  $\sigma$  is substantially more informative about the realised error and supports the publication decision — though, as the text reports, it remains weak as a fine-grained ranking. The dotted line is the publication gate.

TABLE III  
THE TWO COMPONENTS, MEASURED ON REAL DEPTH AGAINST  
ANNOTATED WALLS.

|                                               | median (m) | p90    |
|-----------------------------------------------|------------|--------|
| scatter about the fitted wall (averages down) | 0.0149     | 0.0382 |
| fitted vs. annotated wall (does not)          | 0.0181     | 0.0572 |

here turns on the value: sweeping it, the median error of published corners is flat at 1.99 cm to 2.04 cm for gates of 4, 6, 8 and 10 cm, while the withheld median rises monotonically from 3.09 cm to 5.33 cm and the published fraction from 81.1 % to 93.2 %. The separation is therefore a property of the ordering, not of where the cut is placed. Below 2 cm the gate is vacuous in the other direction: the smallest declared  $\sigma$  in the corpus is 0.0203 m, so a 2 cm gate publishes nothing.

*The actionable claim.* Publication is gated at  $\sigma \leq 6$  cm. A corner the system declares publishable has a median true error of 2.0 cm and lies within the 6 cm it claims in 86.1 % of cases; a corner it withholds has a median error of 3.6 cm and meets the same criterion only 63.7 % of the time. The gate discriminates, and the number it gates on now means something.

### C. The split on recorded depth

The decomposition in (2) claims that error separates into a part that averages away and a part that does not. We measure both on MP3D-FPE [22], which provides real Matterport depth with camera trajectories and per-room annotated corners. Slicing the depth to a horizontal band and fitting each annotated wall from the returns that land on it gives both terms directly (588 segments, 120 rooms; the slice registers to the annotations with a median modal offset of 7.5 mm).

Two consequences. First, the simulated sensor model is consistent with, and slightly more conservative than, the real-depth measurements: real depth scatters at 1.5 cm against the 2.0 cm simulated, so the calibration above rests on a noise model that does not flatter it. Second, the systematic component is not explained by independent sensor noise alone: it survives at 1.8 cm in data whose per-return scatter is 1.5 cm, and scatter that averages down cannot produce an offset that does not. That is what sets up the next result.

### D. Closed-loop simulation with a LiDAR sensor model

The experiments so far establish that the declared uncertainty is calibrated (§V-B) and that its two components are real on real depth (§V-C). Neither shows the estimator *using* them while it runs. This experiment does, on a robot driving a simulated LiDAR in closed loop, in a room whose ground truth is exact.

The room is a rectangle of 6.000 m  $\times$  4.000 m, taken from the wall solids of the world rather than from any fitted model, with a 1.06 m doorway in one wall. A rectangle is the useful test here for a reason that is methodological rather than geometric: *edge lengths and corner angles are invariant to the map frame*, so the estimate can be scored with no rigid alignment. Every other layout comparison in this literature must align first, and alignment absorbs the global pose error into the score. Here it does not arise. We pre-registered the predictions before the run: the polygon closes on four corners, the recovered width falls within two declared  $\sigma$  of truth, and—the outcome we were actually watching—if the pose channel’s known defect, a covariance that omits the repeatable part of the error (§V-E), extends to the layout channel, the declared  $\sigma$  will fall below the realised error while the robot is stationary and keep falling the longer it sits.

TABLE IV

CLOSED LOOP, WEBOTS, LiDAR SENSOR MODEL. GROUND TRUTH IS THE WORLD’S WALL SOLIDS;  $\sigma$  IS THE WORST CORNER OF THE PUBLISHED POLYGON. THE RUN IS 762 s LONG; EACH COLUMN IS LABELLED WITH ITS TIME.

|                       | 0 s     | 665 s   | 689 s          |
|-----------------------|---------|---------|----------------|
| long edge             | 5.994 m | 5.995 m | 5.994 m        |
| short edge            | 4.016 m | 4.013 m | 4.019 m        |
| worst corner $\sigma$ | 0.988 m | 0.989 m | <b>0.048 m</b> |
| worst edge $\sigma_d$ | 0.500 m | 0.500 m | 0.019 m        |

It did not. Table IV reports the run: 665 s parked, then a turn in place.

*The geometry was already accurate before the uncertainty became small.* Standing still, the estimate was already 5.994 m  $\times$  4.016 m—within 6 mm and 16 mm of truth—while the worst corner declared almost a metre of uncertainty and every edge offset sat at exactly its 0.500 m prior. Over 665 s and 13 000 frames, with five wall segments associating per frame, the worst corner  $\sigma$  moved from 0.9882 m to 0.9885 m and the edge prior did not move at all (four of 53 284 logged edge values departed from 0.500 m). A turn in place, over the next 24.5 s, then dropped the worst corner  $\sigma$  by a factor of 21, to 0.048 m, while the geometry moved by millimetres. We do not quote the size of that turn: this run logs the room model’s pose, not the robot’s, so the rotation it performed is not recoverable from it. The third column is the *first* frame the layout is publishable;  $\sigma$  keeps falling afterwards, to 0.031 m at 719 s and 0.029 m at the end of the run, but the log is sampled at 50 ms only up to that frame and has a 30 s gap immediately after it, so the turn is quoted to where it can be measured rather than to where it looks largest.

Two things this run cannot show. The map frame is *gauge-free*—its origin and orientation are wherever the estimator put them—and the log carries no ground-truth robot pose, so we report the no heading error at all, which would describe our own gauge rather than the robot. For the same reason both panels of Fig. 1 are drawn with the long axis horizontal: the re-anchor rotates the frame between them, and every quantity claimed here—edge lengths, corner  $\sigma$ , the comparison with truth—is invariant to that rotation.

This is the generative model’s own arithmetic rather than a policy. A stationary robot observes the same walls from the same viewpoint; the observations are not independent, so the information they carry about the layout saturates. The estimator therefore cannot earn precision by waiting, and does not claim any. Rotation changes the viewpoint, the observations stop repeating, and the precision arrives at once. The final estimate is 5.996 m and 4.005 m against 6.000 m and 4.000 m: errors of  $-4$  mm and 5 mm, or 0.13 and 0.16 of the  $\sigma$  the system declared for them.

The same run shows the opposite behaviour in the pose channel, where correlated evidence does shrink the covariance while the robot is parked. We report that as a measured defect in §VI: it is the same question asked of two channels, and only one of them answers it correctly, which is what makes

the comparison informative.

#### E. Why a single global precision is insufficient

In the same simulator, driving a 166 m tour of a 20-corner apartment, we log for every corner candidate at the association gate the innovation and each term of its covariance: 300 007 candidates over 23 551 frames, of which the association gate rejects 27 %. The tour length is measured from the logged trajectory rather than estimated. The reference layout here is traced from the building’s floor plan rather than read from the world, because that world’s walls are meshes rather than parametrised solids; it is therefore an imperfect map of a kind a deployed robot is often given, which is the point. Normalised innovation squared (NIS, divided by its two degrees of freedom) equals 1 when a covariance matches the error it claims.

A rigid fit between the detections of consecutive frames recovers the inter-frame motion with no pose log, so the population can be split by what the robot is *doing*. Over 34 stationary stretches, which fall in 23 physical stops because a single moving frame ends a stretch, and 386 corner-stretch pairs, decomposing the innovation for a fixed corner separates two parts cleanly: the part that *repeats* has a median of 0.0574 m and the part that *scatters* 0.0066 m, a ratio of 8.7, with the bias exceeding the scatter in 97 % of pairs. Averaging removes the second and cannot touch the first.

*The repeatable part is not a map error.* A map offset displaces the true corner, so the detection re-expressed in the room frame must land in the same wrong place from every viewpoint. It does not: across the 17 corners seen in three or more separate stretches, the spread *between* stretches is 0.0442 m against 0.0074 m *within* one—a factor of 6.0, and of the same order as the bias itself. It is also not sensor noise, which is the part that averages. It is a repeatable, viewpoint-dependent error that the model represents nowhere, and standing still accumulates confidence against it from observations that carry no new information about it.

*The declared precision does not cover this.* Over the same stationary stretches the pose-induced term  $\sigma_{\text{pred}}$  has a median of 0.046 m against a repeatable innovation of 0.057 m: the pose covariance carries no term the size of the bias. The NIS this produces depends on which population is asked, and we report both rather than choosing: over every candidate it is 2.51 at rest and 2.28 in motion, and over the 73 % the association gate accepts, 0.75 and 0.80. Neither sits at 1; the total is too large before the gate and too small after it, and in both cases its pose term is smaller than the bias it must cover.

*How much of this is one run.* The magnitude is run-dependent, since each tour stops at different viewpoints and that is exactly what a viewpoint-attached error predicts; the hardware run below is a second point. What is stable is the shape, and that is what the argument uses.

*The same measurement on a physical robot.* Everything above is simulated. We ran the same pipeline and the same analysis script, unmodified, on a differential-drive robot carrying a 3-D LiDAR in an apartment-scale indoor space, localised against a layout obtained offline by a SLAM system—the

TABLE V  
THE REPEAT/SCATTER SPLIT, SIMULATED AND ON HARDWARE

| tour           | stops   | pairs | repeats  | scatters | ratio        | viewpoint                    |
|----------------|---------|-------|----------|----------|--------------|------------------------------|
| simulated      | 23 (34) | 386   | 0.0574 m | 0.0066 m | $8.7\times$  | $6.0\times$ (17)             |
| physical robot | 3 (6)   | 46    | 0.0624 m | 0.0046 m | $13.4\times$ | $5.9\times$ (8) <sup>†</sup> |

*stops* are physical stops, with the stationary stretches they contain in brackets; pairs, repeats, scatters and ratio are per stretch, as pre-registered. *viewpoint* is the spread between stretches divided by the spread within one, over the corners seen in three or more (count in brackets); above 2 it rules out a fixed map offset. <sup>†</sup>Post hoc, by stop, over corners seen at two or more of the three; per stretch it reads  $9.9\times$  (10), but those stretches are not separate viewpoints.

imperfect map of the kind argued for above—in three sessions, with the estimator restarted between them (Table V). Over 6 stationary stretches and 46 corner-stretch pairs the repeating part has a median of 0.0624 m against 0.0046 m that scatters, with the bias exceeding the scatter in all pairs but one. The viewpoint test is weaker than its numbers suggest. The six stretches are three physical stops, two of them split by a single moving frame, so grouped by stop only one corner is seen from three or more and the pre-registered test is not supported by this run. Relaxed, post hoc, to corners seen from two stops, 8 corners give a spread *between* stops of 0.0607 m against 0.0102 m *within* one, a factor of 5.9. Over accepted candidates NIS is 0.836, against 0.782 in simulation.

Both verdicts point the way the simulator’s do, and the replication is in the bias: 0.0624 m against 0.0574 m, agreeing to 5 mm on a different sensor and a different map. The scattering part is smaller on hardware, in the direction §V-C reports for a simulated sensor deliberately set more conservative than real depth, so the ratio—13.4 against 8.7—exceeds the simulated one for a reason that lives in its denominator rather than in the systematic term the argument is about.

The real sample is much thinner than the simulated one—46 pairs against 386, three stops against 23—and we claim no more than agreement in shape. Its stationary time, about 36 s, is also below the 60 s our pre-registration required for any motion-state claim, so none is made on hardware. The protocol called for deliberate stops at distinct places and orientations; the second session made none, and the first session’s three stretches, which are one stop, gave a between-stretch statistic of 1.6: from a single place the bias repeats, as a viewpoint-attached error predicts. The third session added two short stops, and nothing was collected after it. We report the sequence because it is the statistic’s own sensitivity: a viewpoint-attached error cannot be detected without viewpoint diversity, and distance driven is no substitute for it.

A bias that repeats, follows the viewpoint in simulation and keeps its size on hardware is the argument for representing the offset as an inferred state with a prior rather than a compiled-in number.

#### F. Calibration as a diagnostic

Requiring precisions to mean what they say found three defects of one kind—uncertainty that averages down past an error that does not, in three distinct places. In the *layout* corner

$\sigma$  of Table II the systematic term was absent. In the *corner detector*’s covariance it was present twice under two names for one quantity, making that channel  $2.1\times$  too timid; removing the duplicate and measuring the survivor moved NIS from 0.47 to 1.03 over accepted candidates on the tour it was measured on. In the *pose* covariance feeding  $\sigma_{\text{pred}}$  it is absent again, and was found by an instrument pointed at the layout. We regard the third as the strongest evidence that the formulation does work rather than description: nothing in a pipeline of separately-tuned heuristics depends on a precision being right, so nothing in it would have flagged any of the three.

## VI. LIMITATIONS

Observations in §V-A and §V-B are simulated from the annotated polygons; §V-C validates the noise model but not the full pipeline on recorded data, and the hardware run of §V-E measures one channel rather than closing the loop. Layout accuracy has not been measured on a physical robot, and doing so needs a surveyed reference we do not have. We attempted a full pipeline evaluation on MP3D-FPE [22] and report a negative result we believe is useful to others. Because it annotates *rooms inside open-plan buildings* while a  $360^\circ$  depth sensor sees through every doorway, a median 57 % of returns fall outside the room being scored (0 of its 401 rooms fall below 10 %); taking the union of a floor’s rooms still leaves 50 % outside, because corridors and stairwells are unannotated. No granularity makes the annotation and the observation describe the same region, so this corpus—and, we expect, any room-annotated building corpus—cannot grade a method that models a *bounded* room. The 0.14 IoU the corpus yields under these conditions measures that mismatch rather than the estimator, and we report the mismatch instead. Finally, the pose-covariance defect of §V-E is measured but not fixed; the cure we argue for—a per-wall offset state with the measured prior—is future work.

## VII. CONCLUSION

A layout estimate that a robot can act on needs an uncertainty a robot can check. We have shown that such an uncertainty can be maintained online, that it can be validated against ground truth at scale, and that it becomes useful once something downstream depends on it. Requiring precisions to be calibrated was not bookkeeping: it exposed three defects, and it showed that a quantity commonly treated as a constant is not one.

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responsibility for the content, including all claims and their supporting evidence.

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